

Ivar Cashin Eriksson

Curriculum Vitae

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Introduction

Curious data scientist with seven years' experience applying machine learning across energy, manufacturing, pharmaceutical analytics, healthcare, and e-commerce. I bring a strong statistical background and hands-on experience with forecasting, production ML systems, probabilistic modelling, explainable AI, and stakeholder-facing model delivery.

Selected Work Experience

Full-Stack Data Scientist & Project Manager @ Signum Life Science

- Building forecasting systems, including a pharmaceutical price forecasting model for Denmark's fortnightly blind auctions on an end-to-end Databricks forecasting platform designed for configurable features, probabilistic forecasts, and thousands of simultaneously running models.
- Owned a next-best-action engine and extended explainable AI features with UX designers.

Keywords: probabilistic forecasting, MLflow, Databricks, XGBoost, ML engineering, explainable AI, pricing, pharmaceutical analytics

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Principal Data Scientist @ Valcon

- Built an in-production, real-time, deep-learning system using natural-language inputs to quality control invoicing. Not a single missed invoice in the first week of production.
- Led data science projects across energy, manufacturing, pharma, and IoT.
- Mentored colleagues and led trainings in explainable AI and Git.

Keywords: production ML, neural networks, NLP, explainable AI, energy, stakeholder communication

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Researcher within Data Science @ RaySearch Laboratories

- Built probabilistic ML methods for automated radiotherapy planning.
- Used autoencoders, sparse Gaussian processes, and optimisation on segmented 3D CT scans.
- Collaborated with clinicians and medical physicists on workflow fit.

Keywords: probabilistic modelling, optimisation, deep learning, medical AI, clinical decision support

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Education

- 2015–2020 **Royal Institute of Technology, KTH – Data Science and Engineering**, GPA: 5.0 (out of 5.0)
Master: Statistical Learning and Data Analytics, Bachelor: Engineering Physics. [See thesis](#).
Relevant coursework: Statistical Machine Learning; Applied Time Series; Optimization.
- 2019–2019 **Swiss Federal Institute of Technology, ETH – Applied Mathematics**, Exchange
- 2017–2022 **Stockholm University, SU – Economics**, Supplementary Bachelor's Degree

Skills and Other

- Technology Python, SQL, PyTorch, TensorFlow, Scikit-learn, SHAP, XGBoost, Facebook Prophet, MLflow, Databricks, AWS, MS Azure, FastAPI, Docker.
- Methods Time series forecasting, probabilistic modelling, clustering, explainable AI, autoencoders, CNNs, vector search.
- Language Native: English, Swedish. Intermediate: Danish.

Detailed Work Experience

2024–Present **Full-Stack Data Scientist and Project Manager**, *Signum Life Science*, Copenhagen

At Signum, my current work is on an end-to-end Databricks forecasting platform for pharmaceutical price and demand forecasting, including patient population forecasts. The role spans architecture, project management, data science, ML engineering, and software engineering.

I also maintain and improve a pharmaceutical price forecasting model for Denmark's fortnightly blind auctions. After stabilising the XGBoost-based system, I am now driving efforts toward a more robust and modern implementation.

Earlier in the role, I owned and extended Kwarts, a next-best-action prediction system for pharmaceutical sales representatives. Originally built externally, the system was extended with explainable AI features in collaboration with UX designers, and supported commercialisation by aiding sales representatives. Kwarts adapts to varied customer data maturity, from rule-based logic to deep learning, and is deployed in AWS for scalability.

2022–2024 **Principal Data Scientist**, *Valcon*, Copenhagen

Led data science projects across pharma, energy, IoT, and manufacturing. Mentored junior colleagues in software architecture and data science best practices, led explainable AI and Git trainings, and regularly presented complex technical topics to non-technical stakeholders.

One project involved training a real-time deep neural network with a custom training pipeline and modular architecture. The training data consisted of natural-language inputs and high-dimensional multi-class labels, which made for a challenging prediction problem. I was the lead data scientist and architect of the solution and responsible for implementing the end-to-end production pipeline.

I was awarded the "Valcon People's Choice Award" for exemplary work, leadership, team spirit, and scientific curiosity.

2021–2022 **Data Scientist**, *Valtech*, Copenhagen

Delivered ML solutions in e-commerce. Built time series forecasts using Facebook Prophet, and applied clustering to behavioural data to support personalisation. Worked closely with business stakeholders to ensure actionable outputs.

2019–2021 **Researcher within Data Science**, *RaySearch Laboratories*, Stockholm

At RaySearch Laboratories, I worked on automating radiotherapy planning using a combination of deep learning, probabilistic modelling, and multi-objective optimisation. I developed a pipeline that extracted latent anatomical features from segmented 3D CT scans using autoencoders, and used these representations to identify similar patients via a learned similarity metric. Dose-volume histograms and spatial dose distributions were then predicted using sparse Gaussian processes, providing a probabilistic framework for dose planning conditioned on patient geometry.

Alongside this, I redesigned some of RaySearch's lexicographic optimisation algorithms to better reflect clinical decision hierarchies, particularly in palliative care where trade-offs between tumour control and organ sparing are nuanced. I collaborated closely with medical physicists and clinicians throughout, ensuring the models aligned with oncological standards and practical workflow requirements.

Other Detailed Experience

2025–2025 **Solo Project**, *Project Iris*

In this personal project I am building an end-to-end computer vision system for making product images on e-commerce sites interactive. The pipeline detects individual items using a YOLOv8 model, and embeds them semantically using OpenCLIP. The embeddings are then compared to support intelligent product linking across all images and products in a web store.

The system includes a full-stack setup with a FastAPI backend, MongoDB for persistence, a Qdrant index for vector search, and a custom-built Chromium plug-in that overlays clickable links directly onto web shop imagery in real time. The project also includes a dynamic scraping and indexing pipeline for ingesting and structuring product data from arbitrary e-commerce sites.

The project has been a learning opportunity for fine-tuning of deep learning models and practical ML engineering.